

The Icosahedral Carrier and the D_4 Lattice

Galois Structure, Reduction Modulo 3 and the Doubling $3 \oplus 3'$ —
an Exact Comparison with $T^4/\mathbb{Z}_3$

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14 September 2026

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- The sector pairing $k \leftrightarrow -k$ on $\text{GF}(27)^*$ is the inversion $g^k \mapsto g^{-k}$, not the action of the Frobenius $x \mapsto x^3$ (previously "is its action", "Frobenius on $\text{GF}(27)^*$ "); the identity with $T_1 \leftrightarrow T_2$ holds for the Frobenius on $\text{GF}(9)$.

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Abstract

An externally proposed carrier model for emergent physics uses a twelve-port icosahedron with symmetry group A_5 and a local averaging repair rule on its thirty edges. This document compares that carrier exactly with the FFGFT geometry $T^4/\mathbb{Z}_3$ on the D_4 lattice. Four verification scripts establish: (i) the carrier spectrum lies in $\mathbb{Q}(\sqrt{5})$, and the Galois conjugation $\sqrt{5} \mapsto -\sqrt{5}$ exchanges the two three-dimensional A_5 blocks $3 \leftrightarrow 3'$ as spectral projectors, so that over $\mathbb{Q}$ only the 6D block exists – the doubling $6 = 3 \oplus 3'$ of Dok. 285; (ii) the reduction $\mathbb{Z}[\varphi]/(3) = GF(9)$ turns this Galois conjugation into the Frobenius $x \mapsto x^3$ of the FFGFT ground field; (iii) A_5 acts faithfully and absolutely irreducibly on $GF(9)^3$ but not on $GF(9)^2$, hence not in $G(3)$ but in $G(6)$ – the same threshold as the lepton order 26 in Dok. 341; (iv) A_5 is incompatible with the D_4 lattice ($5 \nmid 1152$, Dok. 314); the common core is the tetrahedral group A_4 , which acts simply transitively on the twelve ports, and the port $\mathbb{Z}_3$ is provably never one of the sixteen orbifold generators of $T^4/\mathbb{Z}_3$. The gauge assignment of the external model ($u(1) \oplus su(2) \oplus su(3)$ on $1 \oplus 3 \oplus (3' \oplus 5)$) is representation-theoretically consistent; its Lie structure is an axiom there, not a consequence of the icosahedron.

List of symbols

The table explains symbols appearing in more than one section. Quantities used only locally are introduced there.

Symbol	Meaning
<i>Graph and carrier</i>	
n	number of ports; $n = 12$ (icosahedron)
A	adjacency matrix; $A_{ij} = 1$ if seam (i, j)
$L = 5I - A$	Laplacian; 5-regular graph
$W = A/5$	averaging operator (expected single-seam update)
μ	Laplacian eigenvalue; $\mu \in \{0, 5-\sqrt{5}, 6, 5+\sqrt{5}\}$
<i>Groups and representations</i>	
A_5	icosahedral rotation group; $ A_5 = 60$
A, T_1, T_2, H	irreducible A_5 representations; dimensions 1, 3, 3, 5
$\varphi = (1+\sqrt{5})/2$	golden ratio
A_4	tetrahedral group; integral elements of A_5 ; $ A_4 = 12$
$\text{Aut}(D_4)$	automorphism group of the D_4 lattice; $ \text{Aut}(D_4) = 1152$
$W(F_4)$	Weyl group of F_4 ; isomorphic to $\text{Aut}(D_4)$
<i>Galois and fields</i>	
$\mathbb{Q}(\sqrt{5})$	number field; generated by the carrier spectrum
$GF(3^k)$	finite field of characteristic 3
$\sigma : \sqrt{5} \mapsto -\sqrt{5}$	Galois automorphism of $\mathbb{Q}(\sqrt{5})$
$x \mapsto x^3$	Frobenius automorphism of $GF(9)$
ζ_n	primitive n -th root of unity
<i>D_4 lattice</i>	
D_4	root lattice; $\{k \in \mathbb{Z}^4 : \sum k_i \equiv 0 \pmod{2}\}$; kissing number 24
$P_{T_1}, P_{T_2}, P_A, P_H$	spectral projectors of L

Status markers

- [B]** Algebraically proven (verification script, exact arithmetic).
- [K]** Numerically confirmed.
- [S]** Structurally plausible, physical identification open.
- [E]** Established external mathematics.
- [X]** Checked and negative.

This document contains no new mathematics. The character table of A_5 , the Galois theory of $\mathbb{Q}(\sqrt{5})$, Brauer degrees, $\text{Aut}(D_4)$, the crystallographic restriction and cut-and-project are textbook material **[E]**. The marker **[B]** here denotes the exact verification of such results on the concrete carrier and their assembly; only the application to the relation between the two frameworks is new.

1 Orientation: geometry, algebra, and what a field is here

FFGFT has two layers that must not be confused.

The geometry is a torus. The carrier is T^4 with the D_4 lattice (kissing number 24, Dok. 314), and the physics sits in the *modes* of this torus: the winding numbers of the plane waves living on it. The built-in $\mathbb{Z}_3$ symmetry of the lattice turns it into the orbifold $T^4/\mathbb{Z}_3$ with nine fixed points. What a particle is, which mass it carries, which sector it belongs to – these are statements about modes and their $\mathbb{Z}_3$ orbits.

The algebra consists of finite fields: $GF(9)$ as ground field, $GF(27)$ with its 26 units in eight three-orbits, both sitting in parallel inside $GF(729)$ (Dok. 338, 341). The Frobenius $x \mapsto x^3$ is the Galois group of these fields; the sector pairing $k \leftrightarrow -k$ is the inversion $g^k \mapsto g^{-k}$. *(Corrected on 1 Oct. 2026: previously “the sector pairing $k \leftrightarrow -k$ is its action” – on $GF(27)^*$, $x \mapsto x^3$ acts as $k \mapsto 3k \bmod 26$, and $3k \equiv -k$ only for $k = 0, 13$; the Frobenius has order 3, the inversion order 2.)*

How the layers connect. The fields *are* not the modes. They are the eigenvalue fields, splitting fields and coefficient fields in which the mode operators live, and they decide which modes *can exist*: a mode is an element order in a matrix algebra over $GF(3^k)$, and Lagrange permits only orders dividing the group order $|GF(3^k)^*|$. That is why order 5 exists in $G(3)$ ($5 \mid 80$) while order 26 appears only in $G(6)$ ($13 \nmid 80$, Dok. 341). In short: *the geometry supplies the modes; the Galois structure of the fields says which of them are admissible*. The fields are mode-conditioned in exactly this sense – they are called up by the modes, not the other way round.

Why this organises the comparison. OPH has the same two-layer structure with different entries: the geometry is a sphere of icosahedra, the modes are the blocks of the port space under A_5 , and the algebra is the number field $\mathbb{Q}(\sqrt{5})$ with its Galois group $\sqrt{5} \mapsto -\sqrt{5}$. There too the Galois structure decides which modes exist over the ground field: over $\mathbb{Q}$ only the 6D block, never a single triplet. The comparison in this document is therefore a comparison of the *algebras that classify the modes* – and the two algebras touch exactly once: in the reduction modulo 3, which maps $\mathbb{Q}(\sqrt{5})$ onto $GF(9)$ and the Galois conjugation onto the Frobenius (Theorem 5.2). The geometries, by contrast, touch only in the tetrahedral group and part at the five-fold symmetry (Section 6).

2 Occasion and scope

In September 2026 a carrier model was presented on the IPI list (Observer Patch Holography, hereafter *OPH*) whose hardware is a finite network of twelve-port icosahedra and whose software is a single local rule: on each of the thirty edges (*seams*) the two port readings are replaced by their arithmetic mean. The author supplied a reproducibility bundle in exact rational arithmetic (300 updates, seeds 7 and 11, an independent verifier).

This document takes no position on the physical claims of OPH. It answers three narrow, exactly checkable questions:

1. What is the structure of the carrier spectrum, and what does its Galois group imply for the distinguishability of the two three-dimensional blocks?
2. How does $\mathbb{Q}(\sqrt{5})$ relate to the FFGFT ground field $GF(9)$?
3. Which part of the icosahedral group A_5 is compatible with the D_4 lattice and the orbifold $T^4/\mathbb{Z}_3$?

The FFGFT reference points are the doubling $6 = 3 \oplus 3'$ (Dok. 285), the embedding $C_3 < A_5$ (Dok. 293), the negative result $5 \nmid |\text{Aut}(D_4)|$ (Dok. 314) and the $G(3)/G(6)$ threshold (Dok. 341).

3 The spectrum of the icosahedral carrier

Theorem 3.1 (Spectrum, **[E]/[B]**). *Let A be the adjacency matrix of the icosahedral graph (12 vertices, 30 edges, 5-regular) and $L = 5I - A$ its Laplacian. Then*

$$\text{spec}(A) = \{5^{(1)}, \sqrt{5}^{(3)}, -1^{(5)}, -\sqrt{5}^{(3)}\}, \quad \text{spec}(L) = \{0^{(1)}, (5 - \sqrt{5})^{(3)}, 6^{(5)}, (5 + \sqrt{5})^{(3)}\}.$$

The only rational non-trivial Laplacian eigenvalue is 6 (multiplicity 5). The field generated by the spectrum is $\mathbb{Q}(\sqrt{5})$, of degree 2.

Proof. Script `spectrum_icosahedral_carrier.py`, SymPy, exact; the seam list is taken from the OPH bundle, alternatively from the canonical vertices $(0, \pm 1, \pm \varphi)$ cyclically. The script also replays the 300 updates of seed 7 in rational arithmetic and reproduces the centered residual recorded in the bundle exactly.

The multiplicities $(1, 3, 5, 3)$ are the dimensions of the irreducible A_5 representations $A \oplus T_1 \oplus H \oplus T_2$ (icosahedral-group notation; H is five-dimensional, T_1, T_2 are the two inequivalent triplets). On a five-fold rotation g_5 they carry the character values

$$\chi_{T_1}(g_5) = \varphi, \quad \chi_{T_2}(g_5) = 1 - \varphi, \quad \chi_{T_1} + \chi_{T_2} = 1.$$

Remark 3.2 (Damping rates, **[E]/[B]**). The expected action of a random single-seam update is $E[M] = I - L/60$ with block eigenvalues $(55 + \sqrt{5})/60 \approx 0.954$ on T_1 , $9/10$ on H and $(55 - \sqrt{5})/60 \approx 0.879$ on T_2 . T_1 is the slowest block; its distance to T_2 is $\sqrt{5}/30$ (ratio ≈ 1.085). This is a statement about rates, not a symmetry selection.

4 Galois conjugation and the doubling $6 = 3 \oplus 3'$

Theorem 4.1 (Galois exchanges $T_1 \leftrightarrow T_2$, **[E]/[B]**). *Let $P_{T_1}, P_{T_2}, P_A, P_H$ be the spectral projectors of L (polynomials in L with coefficients in $\mathbb{Q}(\sqrt{5})$). The Galois automorphism $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ of $\mathbb{Q}(\sqrt{5})$ acts entrywise as*

$$\sigma(P_{T_1}) = P_{T_2}, \quad \sigma(P_{T_2}) = P_{T_1}, \quad \sigma(P_A) = P_A, \quad \sigma(P_H) = P_H.$$

The projector $P_{T_1} + P_{T_2}$ is $\mathbb{Q}$ -rational; P_{T_1} and P_{T_2} separately are not. The minimal polynomial of the T_1 eigenvalue over $\mathbb{Q}$ is $x^2 - 10x + 20$ with roots $5 \mp \sqrt{5}$.

Proof. Script `galois_z3_icosahedral_carrier.py`. The projectors are built as Lagrange polynomials in L ; the exchange is checked by substituting $\sqrt{5} \rightarrow -\sqrt{5}$ and comparing matrices.

Corollary 4.2 (Indivisibility over $\mathbb{Q}$, **[B]**). *Every A_5 -invariant structure on the port space that is defined over $\mathbb{Q}$ as an abstract field – a lattice, an integer character – sees $T_1 \oplus T_2$ as one six-dimensional block; $T_1 \oplus T_2$ is the only six-dimensional submodule with integer character that contains a triplet ($1 \oplus H$ is the other six-dimensional submodule with integer character). Distinguishing T_1 from T_2 is not a symmetry statement.*

Remark 4.3 (The selection happens at the archimedean place, **[B]**). Argument of the author of the external model (personal communication, 15 September 2026), verified exactly here. The repair dynamics does not live in the abstract field $\mathbb{Q}$ but in $\mathbb{R}$: $I - L/60$ is a rational symmetric matrix with real eigenvalues, and the two conjugate rates $(55 \pm \sqrt{5})/60$ are the real numbers 0.954 and 0.879. Both embeddings $\mathbb{Q}(\sqrt{5}) \hookrightarrow \mathbb{R}$ yield the same two subspaces; they differ only in which one carries the name T_1 . Iterating $I - L/60$ on any rational vector with a

component in the slow block, the ratios of the centered norms – a rational sequence – tend to $((55 + \sqrt{5})/60)^2 = 0.9100$ and not to the conjugate value 0.7733 (script, section 6). Selecting the slow triplet as space is therefore a derivation from the dynamics, as every convergent rational sequence selects the positive root of $x^2 = 2$; it happens at the archimedean place of $\mathbb{Q}(\sqrt{5})$. Geometrically: in the coordinates $(0, \pm 1, \pm \varphi)$ the five seam neighbours of a port sum to $\sqrt{5} v$, the thirty second neighbours (edges of the great icosahedron) to $-\sqrt{5} v$; “ T_1 is space” says that repair couples nearest neighbours.

This is the theorem behind Dok. 285: there the doubling $6 = 3 \oplus 3'$ is the answer to the crystallographic restriction (no icosahedral lattice in three dimensions, but one in six). Here the same statement appears as Galois indivisibility at the *finite* places; the archimedean place separates the triplets, the place 3 does not (Theorem 5.2, Section 8). OPH’s proposal that the three-dimensional continuum is the dense projection of “integer frames built from port labels” is the quasicrystal projection $\mathbb{Z}^6 \rightarrow \mathbb{R}^3$; Dok. 285 describes the cause, OPH the image.

Theorem 4.4 (Restriction to $C_3 < A_5$, **[E]/[B]**). *Under a three-fold rotation $g_3 \in A_5$ the port space splits into $\mathbb{Z}_3$ eigenspaces of dimensions $4 + 4 + 4$; blockwise*

$$A \rightarrow 1, \quad T_1 \rightarrow 1 \oplus \omega \oplus \omega^2, \quad T_2 \rightarrow 1 \oplus \omega \oplus \omega^2, \quad H \rightarrow 1 \oplus 2\omega \oplus 2\omega^2.$$

The trivial part has dimension 4, the non-trivial part dimension 8.

This is the C_3 -in- A_5 embedding of Dok. 293 in character form. The non-trivial part (dimension 8) is, as an abstract $\mathbb{Z}_3$ -module, two copies of the orbifold representation on $\mathbb{R}^4$; whether more than representation theory is contained in this is settled negatively in Section 6.

5 Reduction modulo 3: $\mathbb{Z}[\varphi]/(3) = GF(9)$

Theorem 5.1 (Inert prime, **[E]/[B]**). *$x^2 - x - 1$ is irreducible over $GF(3)$ ($5 \equiv 2$ is not a square modulo 3). Hence 3 is inert in $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Z}[\varphi]/(3) \cong GF(9)$.*

Theorem 5.2 (Galois becomes Frobenius, **[E]/[B]**). *Under the reduction of Theorem 5.1 the Galois conjugation $\varphi \mapsto 1 - \varphi$ becomes the Frobenius $x \mapsto x^3$ of $GF(9)$: in $GF(9)$ one has exactly $\varphi^3 = 1 - \varphi$.*

Proof. $\varphi^2 = \varphi + 1$, hence $\varphi^3 = 2\varphi + 1 \equiv 1 - \varphi \pmod{3}$. Script `zphi_mod3_a5.py`, sections (1)–(2).

Thus the Frobenius $x \mapsto x^3$ on $GF(9)$ and the exchange $T_1 \leftrightarrow T_2$ of the icosahedral carrier are not merely analogous: modulo 3 they are the same map. The FFGFT sector pairing on $GF(27)^*$ (Dok. 343/363), by contrast, is the inversion $g^k \mapsto g^{-k}$, not the Frobenius. (Corrected on 1 Oct. 2026: previously “Thus the FFGFT sector pairing (Frobenius on $GF(27)^*$, Dok. 343/363) and the exchange $T_1 \leftrightarrow T_2$... are the same map” – in $GF(9)$, $\varphi^3 = 1 - \varphi$ but $\varphi^{-1} = \varphi - 1$; $T_1 \leftrightarrow T_2$ is thus the Frobenius, the sector pairing on $GF(27)^*$ is the inversion, and no power of the Frobenius is the inversion.)

Theorem 5.3 (A_5 over $GF(9)$, **[E]/[B]**). *The sixty rotation matrices of the icosahedron (entries in $\frac{1}{2}\mathbb{Z}[\varphi]$) reduce modulo 3 to sixty distinct matrices in $GL_3(GF(9))$ spanning all of $M_3(GF(9))$: the reduction is faithful and absolutely irreducible. Entrywise Frobenius of the reduced T_1 matrices yields exactly the reduced T_2 matrices. By contrast $GL_2(GF(9))$ (order 5760, searched exhaustively) contains no $(2, 3, 5)$ -generated subgroup of order 60: A_5 does not embed in $GL_2(GF(9))$.*

Corollary 5.4 (The $G(3)/G(6)$ threshold, **[S]**). *With $G(3) \cong M_2(GF(9))^2$ and $G(6) \cong M_8(GF(9))$ (Dok. 341), A_5 is not in the unit group of $G(3)$ but is in that of $G(6)$. This is the same threshold at which Dok. 341 finds the lepton order 26. The reasons are formally different (Brauer degrees $\{1, 3, 3, 4\}$ of A_5 in characteristic 3 versus Lagrange in $GF(81)^*$); whether the coincidence has substance remains open.*

6 A_5 versus $\text{Aut}(D_4)$

Theorem 6.1 (Reproduction of Dok. 314, **[E]/[B]**). *The automorphism group of the D_4 root system, recomputed from the 24 roots, has order 1152 with order distribution $(1, 2, 3, 4, 6, 8, 12) \mapsto (1, 139, 80, 228, 464, 144, 96)$ and no element of order 5. Among the 80 elements of order 3 exactly 16 are fixed-point-free on $\mathbb{R}^4$ ($|\det(1 - A)| = 9$); each acts freely on the 24 roots (eight orbits of three).*

Corollary 6.2 (**[X]**). *A_5 acts on no D_4 lattice: $5 \nmid 1152$. The symmetry group of OPH and the FFGFT lattice structure are incompatible as wholes.*

Theorem 6.3 (The common core is A_4 , **[E]/[B]**). *The elements of A_5 whose matrices in the model $(0, \pm 1, \pm \varphi)$ are integral (no φ) form the tetrahedral group A_4 (orders 1, 2, 3 with counts 1, 3, 8). A_4 acts simply transitively on the twelve ports – the port set is an A_4 -torsor – and embeds as $\text{diag}(M, 1)$ in $W(D_4) < \text{Aut}(D_4)$.*

This sharpens Dok. 285 (" $C_3 < A_5$ "): not only the three-fold symmetry but the whole tetrahedral group of order 12 – the port count – is lattice-compatible; only the five-fold symmetry drops out.

Theorem 6.4 (Ports as half the first shell, **[E]/[B]**). *Every three-fold rotation of A_5 acts freely on the twelve ports (four orbits of three). As a $\mathbb{Z}_3$ -module the port space is $4 \cdot \mathbb{Z}[\mathbb{Z}_3]$, the first D_4 shell is $8 \cdot \mathbb{Z}[\mathbb{Z}_3]$; the ports are one half of " $24 = 12 + 12$ " from Dok. 314.*

Theorem 6.5 (The port $\mathbb{Z}_3$ is not an orbifold $\mathbb{Z}_3$, **[X]**). *The three-fold rotation permuting the ports is a rotation of $\mathbb{R}^3$ and fixes an axis; as $\text{diag}(M, 1)$ it fixes a line in $\mathbb{R}^4$ and is therefore one of the $80 - 16 = 64$ non-orbifold elements of order 3. This is not an artefact of the embedding: for each of the 16 orbifold generators and each of the 70 choices of four of its eight root orbits (1120 $\mathbb{Z}_3$ -stable twelve-element halves) it was checked whether the twelve roots form a cuboctahedron (D_3 half) or carry the icosahedral graph on some inner-product class. Neither ever occurs.*

Proof. A $\mathbb{Z}_3$ stabilising a D_3 half stabilises its normal up to sign, hence exactly for order 3; it has a fixed vector and is not an orbifold generator. Script `a5_vs_d4.py`, sections (E)–(F).

Remark 6.6 (Side result, **[K]**). *256 of the 1120 orbifold-stable twelve-element halves are tight frames in $\mathbb{R}^4$ (Gram eigenvalue 6 with multiplicity 4, the 4D counterpart of the OPH relation $G^2 = 4G$ for the port vectors). Their graph spectra contain $\sqrt{13}$, not $\sqrt{5}$; the stabiliser has order 18 and is not transitive. They carry no port structure.*

7 The OPH gauge assignment

OPH assigns to the port module the Lie algebra $u(1) \oplus su(2) \oplus su(3)$: $u(1)$ on A , $su(2)$ on T_1 , $su(3)$ on $T_2 \oplus H$.

Theorem 7.1 (Consistency of the assignment, **[E]**). A_5 sits in $SO(3) \subset SU(3)$ via T_2 . The adjoint of $su(3)$ is $3 \otimes \bar{3} - 1$; under $SO(3)$ it splits as $3 \oplus 5$, under A_5 with the T_2 embedding as $T_2 \oplus H$ ($\Lambda^2 T_2 = T_2$, $\text{Sym}^2 T_2 = 1 \oplus H$). Likewise the adjoint of $su(2)$ under the T_1 embedding is T_1 . The restriction of $u(1) \oplus su(2) \oplus su(3)$ to A_5 is therefore exactly $A \oplus T_1 \oplus T_2 \oplus H$.

Remark 7.2 (What is axiom, **[S]/[E]**). The derivation of the centre in this remark is due to the author of the external model (personal communication, 15 September 2026). The assignment presupposes two axioms of the external model: (A1) the reversible port response is a compact Lie algebra $\mathfrak{g}$ with a faithful representation on the port module; (A2) A_5 acts on $\mathfrak{g}$ by inner automorphisms. The one-dimensional centre then follows and is not a third axiom: inner automorphisms act trivially on the centre, so it lies in the A_5 -fixed part of $1 \oplus 3 \oplus 3' \oplus 5$, the uniform line. Centre zero would require $\dim \mathfrak{g} = 12$ semisimple, i.e. $su(2)^4$; under an inner action each $su(2)$ factor has fixed dimension 0 or 3, so the fixed dimension would be a multiple of 3, whereas the port module has fixed dimension 1. Hence the centre is the uniform line and $11 = 8 + 3$ is the only decomposition of the rest into dimensions of compact simple algebras. The counterexample $u(1)^3 \oplus su(2)^3$ fails on (A2): under an inner action its module contains no 5; and even without (A2) it fails by Theorem 4.1, since the centre of a torus is defined over $\mathbb{Q}$ and can therefore contain 3 and $3'$ only with equal multiplicity. The correct statement is thus “ A_5 plus (A1), (A2) forces the Lie type”.

FFGFT is in the mirror situation: $su(3)$ follows from the $\mathbb{Z}_3$ triality on T^4 **[B]** (Dok. 321), $su(2)$ is open **[S]** (Dok. 324). In OPH T_1 is clean and $8 = T_2 \oplus H$ is the axiomatic step.

8 Phases as roots of unity: how the connection succeeds

Section 6 and Theorem 5.2 seem to contradict each other: five-fold symmetry is incompatible with the lattice, yet after reduction A_5 sits in $GL_3(GF(9))$. The contradiction dissolves once one distinguishes *as what* a symmetry enters the carrier – as a lattice automorphism or as a phase.

Theorem 8.1 (Phase channel, **[E]/[B]**). 1. Modulo 3, φ is a primitive element of $GF(9)^*$: $\varphi^4 = -1$, order 8.
 2. The 72° rotation has characteristic polynomial $(x - 1)(x^2 - (\varphi - 1)x + 1)$. Modulo 3 the quadratic factor has no root in $GF(9)$; its roots lie in $GF(81)$ and have order 5 there ($5 \mid 80$).
 3. $13 \mid 728 = |GF(729)^*|$ but $13 \nmid 80$: order 26 needs the cubic extension (Dok. 341).
 4. In characteristic 3, $x^3 - 1 = (x - 1)^3$: no $GF(3^k)$ contains a primitive cube root of unity. A $\mathbb{Z}_3$ acts unipotently in the field, never as a phase ω .

Proof. Script `phasen_einheitswurzeln.py`.

Reading. A phase is a root of unity, and roots of unity are exactly what a field extension adds. In the lattice the 72° rotation is an automorphism of vectors, and $5 \nmid 1152$. In the field the same rotation is only its eigenvalue ζ_5 , and $5 \mid 80$. Geometric incompatibility becomes arithmetic compatibility as soon as one passes from lattice vectors to phases. This is the mechanism behind Dok. 293: $\theta = 2/9$ arises from R_5 as a phase, not as a lattice operation.

The extension degrees are the two basic operations of FFGFT:

Extension	Degree	carries	Counterpart
$GF(9) \rightarrow GF(81)$	2	$\zeta_5, \varphi, 2/\sqrt{5}$, icosahedron	doubling $6 = 3 \oplus 3'$ (Dok. 285)
$GF(9) \rightarrow GF(729)$	3	ζ_{13} , order 26, leptons	$\mathbb{Z}_3$ triality (Dok. 341)

And three-fold symmetry itself is *not* a phase: in characteristic 3 ω collapses to 1 (Dok. 341: the N eigenvalues $\{-3, +3\}$ become $\{0\}$). This is why in FFGFT the $\mathbb{Z}_3$ is lattice structure – a circulant on the mode orbits, Dok. 314 – and five-fold symmetry is a phase. It is not a choice but forced: what cannot exist as a root of unity in characteristic 3 must enter as a lattice operation; what does not fit the lattice must enter as a root of unity in an extension. The two channels are complementary, and the prime 3 decides which symmetry falls into which channel. In the language of places of $\mathbb{Q}(\sqrt{5})$: the external model chooses between T_1 and T_2 at the archimedean place (Remark 4.3); FFGFT computes at the place 3, where $\sqrt{5}$ has no sign and the two triplets are exchanged by the Frobenius. Both selections are correct, at different places – and that is why the two frameworks touch exactly once. Theorem 6.5 is the geometric side of the same statement: the port $\mathbb{Z}_3$ is lattice, the orbifold $\mathbb{Z}_3$ is lattice, but they are different lattice operations; the five-fold symmetry that distinguishes them is not present in the lattice at all.

9 What is projected and what is not: scope of the conclusions

The reference object of this document is the torus, not the icosahedron. The icosahedron is *one* geometric body among many that can be projected onto $T^4/\mathbb{Z}_3$, and it is the test body here only because an external model proposes it as carrier. Every other finite body – the five-fold geometry of Dok. 285 was the previous case – splits by the same pattern into a lattice-compatible part (in $\text{Aut}(D_4)$) and a phase part (roots of unity in an extension $GF(3^k)$); the icosahedron is the instance on which the pattern is worked out completely in this document, not the pattern itself. None of the maps is a complete projection of the icosahedron onto the torus; by Theorem 6.1 no such projection can exist, and that holds for every body with non-crystallographic symmetry. In the terminology of Dok. 330 the maps are neither Type I (lossless decompactification) nor Type III (bijective representational translation), but lossy like Type II – or not FFGFT operations at all. The conclusions are justified exactly as far as the respective map is faithful on what is being concluded. They also hold relative to the choice of lattice: D_4 is a posit (Dok. 320 **[POSIT]**), motivated by the densest packing in four dimensions and by the fact that D_4 is the only lattice of the D_n family with S_3 outer symmetry, which carries the $\mathbb{Z}_3$ as triality (Dok. 314; there also the Casimir counter-finding in favour of $\mathbb{Z}^4$). All statements of this document about $\text{Aut}(D_4)$ presuppose that choice.

Genuine projections (faithful on the object of the conclusion).

1. $\mathbb{Z}_3$ on the mode orbits (Dok. 314): intrinsic torus structure, complete. All triality conclusions rest on it.
2. $\mathbb{Z}[\varphi] \rightarrow GF(9)$ (Theorem 5.1): surjective ring homomorphism with kernel (3). Faithful for everything at the place 3 (mode orders, extension degrees, indistinguishability of T_1, T_2); lost is the sign of $\sqrt{5}$ and with it everything archimedean (rates, order, convergence). No conclusion about dynamics.
3. Rotation $\mapsto$ eigenvalue (Theorem 8.1): faithful for the order, with maximal kernel – the geometry is forgotten. The conclusion “five-fold symmetry only as a phase” holds for precisely that reason.
4. $A_4 \hookrightarrow W(D_4)$ (Theorem 6.3): faithful for everything tetrahedral, five-fold symmetry not included.

Not projections of the torus.

5. Ports = half the first shell: isomorphic only as $\mathbb{Z}_3$ -sets; the icosahedral edges do not exist in D_4 . Counting statements are justified, statements about neighbourhood or rates are not.
6. Cut-and-project $\mathbb{Z}^6 \rightarrow \mathbb{R}^3$: a genuine projection of a *different* lattice ($\mathbb{Z}^6/D_6$, the second protected geometry of Dok. 285), not of T^4 .
7. Port $\mathbb{Z}_3 \rightarrow$ orbifold $\mathbb{Z}_3$: provably not a projection (Theorem 6.5).
8. Gauge assignment and choice of space in the external model: statements at the archimedean place about its response algebra; none of it projects onto T^4 .

What kind of confirmation this is. The projections do not confirm that the torus is the physics. They confirm that the torus geometry makes concrete, falsifiable structural predictions and that an independently constructed foreign object satisfies them. Four predictions had the opportunity to fail:

- *Lattice*: order 5 does not act on D_4 (Dok. 314). Had $A_5 \subset \text{Aut}(D_4)$, Dok. 285 would be wrong. Passed.
- *Field*: ζ_5 only in $GF(81)$, ζ_{13} only in $GF(729)$, no ζ_3 (Dok. 341). Had 3 split in $\mathbb{Q}(\sqrt{5})$ or $\zeta_5 \in GF(9)$, the correspondence doubling $\leftrightarrow$ degree 2 would be wrong. Passed.
- *Orbifold*: $\mathbb{Z}_3$ is a lattice operation, not a phase. The port $\mathbb{Z}_3$ could have been an orbifold $\mathbb{Z}_3$; it is not, for a reason that mirrors the torus structure. Passed.
- *Mode order*: the $G(3)/G(6)$ threshold meets the same boundary for A_5 as for order 26. The weakest test, **[S]**.

This is confirmation in the sense of a survived test: more than internal consistency, less than proof – and it applies to the torus, not to the icosahedron. A further test body (octahedron, cube, a four-dimensional polytope) would put the same four predictions to the question again, with a different lattice part and a different number field; the tetrahedral and octahedral groups lie entirely in $\text{Aut}(D_4)$ and would have no phase part, the 600-cell would bring back $\mathbb{Q}(\sqrt{5})$. In this series the icosahedron is only the first fully computed case. Unexamined – and not examinable by any projection – remains whether T^4 is the physical carrier, whether $\theta = 2/9$ is physically real as a phase (still **[K]**, Dok. 293) and whether mode orders are particles; that requires measurement. Equally untouched is the deeper question whether FFGFT's own translations (Type III, Dok. 270/330) are complete: there, internal consistency without any statement about observability continues to hold. Whoever reads more into this document than the location of the single point of contact overreaches.

Tool and method

Claude (Anthropic) was used as a tool for: writing and executing the Python verification scripts; drafting the $\text{\LaTeX}$ sources from results and corrections of the author; maintaining the changelog and cross-references. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. Every algebraic statement is backed by a Python verification script with explicit assertions — if an assertion fails, the statement is not used.

10 Summary

	OPH	FFGFT	Link
Field	$\mathbb{Q}(\sqrt{5})$	$GF(9), GF(27) \parallel GF(729)$	$\mathbb{Z}[\varphi]/(3) = GF(9)$; Galois $\rightarrow$ Frobenius [B]
Group	A_5	$\text{Aut}(D_4), \mathbb{Z}_3$	$A_5 \not\subset \text{Aut}(D_4)$; core A_4 (= 12 ports) [B]
$3 \oplus 3'$	T_1 space, T_2 colour	6D lattice (Dok. 285)	indivisible over $\mathbb{Q}$ [B]
$\mathbb{Z}_3$ orbits	4 free on 12	8 free on 24	ports = half the first shell [B]
Port $\mathbb{Z}_3$	axis in $\mathbb{R}^3$	orbifold fixed-point-free	never conjugate [X]
$G(3)/G(6)$	$A_5 \not\subset GL_2(GF(9)), \subset GL_3$	order 26 only in $G(6)$	same threshold [S]

Verification scripts. 2/python/Dok364_Skripte/: spectrum_icosahedral_carrier.py, galois_z3_icosahedral_carrier.py, zphi_mod3_a5.py, a5_vs_d4.py, phasen_einheitswurzeln.py; batch run run_all_364.sh. The seam list of the external bundle is read if present; otherwise the canonical icosahedron is used. No register entry: no earlier document is corrected.

Bibliography

- [1] J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1993. Standard reference for $\text{Aut}(D_4)$, kissing number 24, order distribution.
- [2] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge University Press, 1990. D_4 triality, Dynkin diagram, $W(F_4)$.
- [3] J.-P. Serre, *Linear Representations of Finite Groups*, Springer, 1977. Character table of A_5 ; Brauer degrees in characteristic 3.
- [4] R. Lidl, H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997. $GF(9)$ as extension of $GF(3)$; Frobenius automorphism; roots of unity in $GF(3^k)$.
- [5] J. Neukirch, *Algebraic Number Theory*, Springer, 1999. Inertness of 3 in $\mathbb{Q}(\sqrt{5})$; Galois group of $\mathbb{Q}(\sqrt{5})/\mathbb{Q}$.
- [6] M. Berger, *Geometry I*, Springer, 1987. Icosahedron, A_5 symmetry; canonical vertices $(0, \pm 1, \pm \varphi)$.
- [7] J. Pascher, *Dok. 285: FFGFT — HLV, CP violation and dimension bridge*, FFGFT corpus.
- [8] J. Pascher, *Dok. 293: Icosahedron, $\theta = 2/9$ and C_3 -in- A_5* , FFGFT corpus.
- [9] J. Pascher, *Dok. 314: Lattice in Hilbert space*, FFGFT corpus.
- [10] J. Pascher, *Dok. 341: $GF(27)$ and GALG in FFGFT*, FFGFT corpus.